A sharp three-point kernel bound and conditional proportions of zeta zeros

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github.com/fsantibanezleal/CAOS_RESEARCH

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Abstract

Wang recently refined Lamzouri's Hilbert-space method by retaining a convex spectral defect and proved that at least 0.6725007703418699... of the nontrivial zeros of the Riemann zeta function are simple and on the critical line. The quantitative input is a three-point lower bound for a limiting Fourier kernel. We determine the sharp constant in the elementary ratio that controls this bound. For all $\alpha, \beta \geq 0$,

$$\frac{\sqrt{(1+\alpha^2)(1+\beta^2)} + \alpha\sqrt{1+\alpha^2} + \beta\sqrt{1+\beta^2}}{1+\alpha^2+\alpha\beta+\beta^2} \leq \sqrt{2},$$

with equality only at $(0,1)$ and $(1,0)$. Substitution into Wang's argument gives the simple-critical proportion

$$0.672500799594675755828355056296394 7\ldots$$

and the distinct-zero proportion $0.836250399797337877914177528148197 3\ldots$. The new correction above the Montgomery–Taylor constant is about 43.88% larger than Wang's correction. We also give a short-interval corollary for the source-certified rank-six parity curve. All numerical comparisons use directed rational enclosures and an independent 120-decimal interval replay. The global transfer and the short-interval corollary assume the framework of Wang's preprint arXiv:2609.24167v1. On 6 October 2026 its author replaced that preprint by a notice stating that a new method is needed to obtain a good improvement. The two proportions above are therefore conditional on an argument that its author no longer offers, and are not claimed here; only the sharp ratio theorem is claimed unconditionally. No effective height, density-one result, universal simplicity statement, or proof of the Riemann hypothesis is claimed.

1 Introduction

Let $N(T)$ count the nontrivial zeros $\rho = \beta + i\gamma$ of the Riemann zeta function with $0 < \gamma \leq T$, including multiplicity. Let $N_0^s(T)$ count the simple zeros among them on $\beta = 1/2$, and let $N_d(T)$

count distinct zeros, each once. Define

$$C_0 = \frac{3}{2} - \frac{1}{\sqrt{2}} \cot(1/\sqrt{2}) = 0.67250070367941164573437979080329 51\ldots. \tag{1}$$

Alpöge and Furman [1], through a proof whose central optimization was found computationally, established that the lower density of simple critical zeros is at least C_0. Lamzouri [3] gave a direct Hilbert-space proof. Wang [6] retained a nonnegative spectral defect in Lamzouri's finite inequality and combined it with disjoint three-point blocks. His result is

$$\liminf_{T\to\infty} \frac{N_0^s(T)}{N(T)} \geq C_0 + 6.66624\ldots \times 10^{-8}.$$

The only change made here to Wang's analytic framework is an exact optimization inside his three-point kernel lemma. Wang obtains an auxiliary ratio and bounds it above by 2. Theorem 2.1 shows that its exact maximum is $\sqrt{2}$. This raises the kernel energy, and all subsequent steps of the global proof remain valid with the new constant.

Our main consequence is the following.

Theorem 1.1. *Assuming the analytic and finite-dimensional framework of Wang [6],*

$$\liminf_{T\to\infty} \frac{N_0^s(T)}{N(T)} \geq 0.67250079959467575582835505629639 47\ldots, \tag{2}$$

$$\liminf_{T\to\infty} \frac{N_d(T)}{N(T)} \geq 0.83625039979733787791417752814819 73\ldots. \tag{3}$$

The displayed decimals are certified lower bounds after downward rounding.

The dependency phrase in Theorem 1.1 is deliberate. The sharp ratio theorem below is self-contained, while the passage to zeta zeros uses a September 2026 v1 preprint that has not been peer reviewed. We inspect the substitution point and reproduce the numerical constants, but do not relabel the imported framework as independently established literature. On 6 October 2026 the author of that preprint replaced it on arXiv by a notice stating that a new method is needed to obtain a good improvement. We have not located the point that prompted the notice; whatever it is, Theorem 1.1 and the corollary of Section 5 now rest on an argument its author no longer offers, and we do not claim either. Theorem 2.1 does not depend on that argument.

2 The sharp auxiliary ratio

For $\alpha, \beta \geq 0$, set

$$R(\alpha,\beta) = \frac{\sqrt{(1+\alpha^2)(1+\beta^2)} + \alpha\sqrt{1+\alpha^2} + \beta\sqrt{1+\beta^2}}{1+\alpha^2+\alpha\beta+\beta^2}. \tag{4}$$

Theorem 2.1 (Sharp ratio theorem)**.** *For every $\alpha, \beta \geq 0$,*

$$R(\alpha,\beta) \leq \sqrt{2}.$$

Equality holds exactly at $(\alpha,\beta) = (0,1)$ and $(1,0)$.

Proof. Write

$$\alpha = \sinh u, \qquad \beta = \sinh v, \qquad u, v \geq 0,$$

and put

$$X = \cosh(u+v), \qquad Y = \cosh(u-v), \qquad S = \sinh(u+v) = \sqrt{X^2-1}.$$

Then $X \geq Y \geq 1$. Product-to-sum identities give

$$\cosh u \cosh v = \frac{X+Y}{2}, \qquad \sinh u \cosh u + \sinh v \cosh v = SY.$$

Hence the numerator of (4) is

$$\frac{X}{2} + Y\left(S + \frac{1}{2}\right). \tag{5}$$

A direct expansion gives the denominator as

$$1 + \sinh^2 u + \sinh u \sinh v + \sinh^2 v = \frac{X}{2} + Y\left(X - \frac{1}{2}\right). \tag{6}$$

Fix $X > 1$. The derivative of the quotient of (5) and (6) with respect to Y has the sign of

$$1 + S - X.$$

This is positive, since $S > X - 1$ follows by squaring from $X^2 - 1 > (X-1)^2$. The quotient is therefore largest at $Y = X$. Equality $Y = X$ means $\cosh(u-v) = \cosh(u+v)$, and for $u, v \geq 0$ this is equivalent to $u = 0$ or $v = 0$.

On this boundary the ratio is

$$\frac{1 + \sqrt{X^2-1}}{X}.$$

All quantities are nonnegative, and comparison with $\sqrt{2}$ reduces, after two legitimate squarings, to

$$4(X^2-1) \leq X^4 \quad \Longleftrightarrow \quad (X^2-2)^2 \geq 0.$$

Equality requires $X = \sqrt{2}$. Together with $u = 0$ or $v = 0$, this gives the two stated points. If $X = 1$, then $u = v = 0$ and $R = 1$. □

Remark 2.2. The parametrization is onto the whole nonnegative quadrant and the argument includes all unbounded directions. The proof is exact; no grid, compactification, or numerical optimizer is used.

3 The improved three-point kernel bound

We recall the exact point at which Theorem 2.1 enters Wang's argument. Let f_0 be the limiting compactly supported cosine profile in [3, 6], let $K_0 = \widehat{f_0}$, and define

$$a = \sqrt{2}\pi \cot(1/\sqrt{2}), \qquad \mathcal{F}(x) = ax \sin(\pi x) - \cos(\pi x).$$

The Fourier transform has the identity

$$\mathcal{F}(x) = (2\pi^2 x^2 - 1) K_0(x). \tag{7}$$

For $u, v \geq 0$, Wang sets

$$d = \max\{|\mathcal{F}(u)|, |\mathcal{F}(v)|, |\mathcal{F}(u+v)|\}, \qquad \alpha = au, \quad \beta = av.$$

His vector triangle estimates and trigonometric addition identity yield

$$(1-d)^2 \leq d(1+d)R(\alpha, \beta) \tag{8}$$

when $d < 1$; the desired lower bound is automatic when $d \geq 1$. Wang then uses $R \leq 2$. Theorem 2.1 instead gives

$$(1-d)^2 \leq \sqrt{2}\, d(1+d).$$

The positive equality root is

$$d_\dagger = \frac{\sqrt{2+8\sqrt{2}} - (2+\sqrt{2})}{2(\sqrt{2}-1)} = 0.28316543080853732700529814807813 29\ldots. \tag{9}$$

Thus $d \geq d_\dagger$. Exact directed enclosures also prove

$$d_\dagger > \frac{\sqrt{57}-7}{2} > \sqrt{5}-2,$$

so the result strictly improves both our initially declared fallback and Wang's constant.

Combining (7) with $u + v \leq H$ gives the sharpened kernel lemma.

Corollary 3.1. *For every $H > 0$ and $u, v \geq 0$ with $u + v \leq H$,*

$$K_0(u)^2 + K_0(v)^2 + K_0(u+v)^2 \geq e_\dagger(H) := \frac{d_\dagger^2}{(1+2\pi^2H^2)^2}. \tag{10}$$

Proof. At least one of the three $\mathcal{F}$ values has modulus at least $d_\dagger$. For its argument x, $|x| \leq H$ and $|2\pi^2x^2 - 1| \leq 1 + 2\pi^2H^2$. Equation (7) therefore makes the corresponding squared kernel value at least the right side of (10). □

4 Global proportions

We give enough of the transfer to identify every dependence and constant. For a conjugation-invariant finite zero multiset $\mathcal{Z}$, let n denote the number of simple real elements and let

$$G_K = (K(x_j - x_\ell))_{1 \leq j, \ell \leq n}$$

be their Gram matrix. Wang defines a nonnegative convex spectral defect

$$\Delta_K(\mathcal{Z}) = \operatorname{tr} \Psi(G_K), \qquad \Psi(t) = \begin{cases} (t-1)^2, & 0 \leq t \leq 2, \\ 2t - 3, & t \geq 2. \end{cases}$$

His finite inequality is

$$n \geq 2|\mathcal{Z}| - \sum_{z,s \in \mathcal{Z}} K(z-s)^2 + \Delta_K(\mathcal{Z}), \tag{11}$$

with multiplicity in the sums. A companion inequality gives the distinct count. His block lemma says that each disjoint simple-real triple whose three squared pair kernels sum to at least e contributes at least $2e$ to the defect.

Partition the scaled ordinate interval into cells of length at most H and form disjoint triples in each cell. At most two simple points per cell are unused. Corollary 3.1 and the Riemann–von Mangoldt asymptotic then give, after the same smoothing and ordered limits as in [6],

$$\liminf_{T\to\infty} \frac{N_0^s(T)}{N(T)} \geq u_\dagger(H) := \frac{C_0 - 2a_\dagger(H)/H}{1 - a_\dagger(H)}, \qquad a_\dagger(H) = \frac{2e_\dagger(H)}{3}. \tag{12}$$

The smoothing parameter remains fixed while T tends to infinity and is removed only afterward. Since (10) is uniform on every fixed cell, the sharper constant does not alter that order of limits.

Choose the rational value

$$H_* = \frac{372019}{100000} = 3.72019. \tag{13}$$

Directed arithmetic gives

$$\begin{aligned} u_\dagger(H_*) - C_0 &= 0.000000095915264110093975265493099597 9\ldots, \\ u_\dagger(H_*) &= 0.672500799594675755828355056296394786 5\ldots. \end{aligned}$$

This proves (2).

For comparison, using Wang's $\sqrt{5} - 2$ and his exact $H_0 = 372018941724/10^{11}$ in the same certificate gives

$$\delta_0 = 0.000000066662458333412524952120753924 1\ldots,$$

strictly between 6.66624×10^{-8} and 6.66625×10^{-8}. The lower endpoint of the new gain interval exceeds the upper endpoint of this reproduced value. Their ratio shows that the new correction is about 43.88% larger. We make no claim that the convenient rational H_* is the exact optimizer.

Wang's distinct-zero inequality, with the same replacement, is

$$\liminf_{T\to\infty} \frac{N_d(T)}{N(T)} \geq \frac{1 + u_\dagger(H)}{2}.$$

Inserting (13) proves (3).

5 A short-interval corollary

The same cell packing can be placed inside the fixed-power intervals used in the companion CAOS manuscript [2]. Let $h_6(\theta)$ be its rank-six Hilbert–parity lower curve, $k_6(\theta)$ the attributed odd-support term, and $q(\theta)$ the pair statistic. If $h_6(\theta) > 0$, choose a fixed cell length $H > 2/h_6(\theta)$ and set

$$\alpha_H = \frac{2e_\dagger(H)}{3}, \qquad \beta_H = \frac{2\alpha_H}{H}.$$

The number of cells divided by the local zero count tends to $1/H$. The triple count and Wang's block lemma therefore give

$$\liminf \frac{\Delta_K}{N} \geq \alpha_H s - \beta_H,$$

where s is the lower simple-critical proportion. Substitution into the finite defect–parity product of [2] yields

$$2(1-s)^2 \leq (1 - k_6(\theta))\{q(\theta) + \beta_H - (1 + \alpha_H)s\}. \tag{14}$$

At $\theta = 0.5459$ and the integer cell length $H = 140730$, the exact certificate proves $H > 2/h_6$. The smaller root J_6 of equality in (14) satisfies

$$J_6 - h_6 > 3.086780998333418751324384201974 19 \times 10^{-31}. \tag{15}$$

The bound is correlated: it follows from the value of the strengthened quadratic at the old root and a derivative bound, rather than subtraction of two independently rounded roots. It exceeds the earlier spectral correction $1.776662254114\ldots \times 10^{-68}$ by more than 37 orders of magnitude. It remains too small to alter the useful decimals of h_6 and does not move the positivity onset.

6 Exact certificate and limitations

The machine-readable record [4] pins both files of arXiv:2609.24167v1, the experiment declaration and amendment, the preceding rank-six certificate, the runner, and the canonical output. The principal calculation uses exact fractions, integer square-root enclosures, and directed Taylor bounds for elementary functions. A separate implementation using 120-decimal `mpmath.iv` intervals overlaps every headline interval. The canonical CPU run completed in 37.157 seconds from a clean Git commit.

The certificate verifies

- the exact ordering of all three kernel constants;
- Wang's printed correction at his rational cell length;
- strict interval separation between the new and old global gains;
- the short-interval cell condition and positive correlated reserve;
- strict dominance over the preceding spectral correction; and
- all pinned source and execution hashes.

Theorem 2.1 is unconditional. The global transfer is conditional on the correctness of the cited Wang v1 argument, although the exact substitution point and downstream algebra were rechecked. The short-interval corollary additionally imports a rank-six constant whose source does not print its coefficient vector. The statements are asymptotic and supply no effective starting height. They neither show that every zero lies on the critical line nor that every zero is simple.

Since 6 October 2026 the cited preprint is replaced on arXiv by its author's notice that a new method is needed to obtain a good improvement. The pinned v1 files still define exactly which argument the conditional statements assume, and the certificate still verifies the algebra downstream of the substitution point, but neither supports the proportions without that argument. Of the results above, only Theorem 2.1 and the kernel-level inequality of the section on the improved three-point bound are claimed. The short-interval positivity onset of the companion manuscript [2] uses a different source, Wang's short-interval pair-correlation theorem (arXiv:2609.07918), whose statement is unchanged in its revised version.

Computational assistance disclosure

ChatGPT-6 Astra was used to search the recent literature, explore the hyperbolic substitution, implement interval certificates, and check the manuscript. The author inspected the primary sources, verified the proof and equality cases, reviewed every stated dependency and limitation, and takes responsibility for the content. No artificial-intelligence system is listed as an author.

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